

Instructor Key: Day 6 Mastery Quiz: Branch Routing

Condition result to selected branch to action evidence

Instructor key. Same layout as student quiz; solutions filled in response boxes. Print duplex on one sheet.

Name		Section		Date	
Score	_____/10		Mastery check		secure / developing / needs support

Engineering situation and quiz rule

The sensor-kit checkout team is deciding whether one kit must be charged before it moves to the next checkout station. Trace the condition, name the branch that runs, and explain the selected action.

1 Reference code for Items 1–10

[trace](#) [explain](#)

```
1 kit_label = "S-21"
2 charge_percent = 76
3 charge_minimum = 80
4 calibration_label = "current"
5 borrower_initials = "RG"
6
7 charge_ok = charge_percent >= charge_minimum
8
9 if charge_ok:
10     action = "send to calibration check"
11 else:
12     action = "charge before checkout"
13
14 route_note = kit_label + ": " + action
```

2 Items 1–5: trace the route evidence

[predict](#) [route](#)

Pt.	Prompt	My response
1	What value is stored in charge_ok? Show the comparison that creates it.	Key: False. The comparison is <code>76 >= 80</code> , which is false.
1	Which branch runs for kit S-21: the if branch or the else branch?	Key: The else branch runs because charge_ok is false.
1	What value is stored in action?	Key: "charge before checkout".
1	What value is stored in route_note?	Key: "S-21: charge before checkout".
1	If charge_percent were exactly 80, what would charge_ok be? Explain the boundary.	Key: True. The comparison uses <code>>=</code> , so <code>80 >= 80</code> passes.

Pt.	Prompt	My response
1	If charge_percent were 82, which branch would run and what action would be assigned?	Key: The if branch runs and action becomes "send to calibration check".
1	Explain why calibration_label does not control this route yet.	Key: calibration_label is stored, but no condition in this code uses it to choose the branch.
1	A teammate says both branches run because both are written in the program. Correct that claim.	Key: Only one branch runs: if the condition is true the if block runs; otherwise the else block runs.
1	Write one complete evidence sentence naming the condition result, branch followed, and selected action.	Key: Secure answer names false condition, else branch, and charge-before-checkout action.
1	What support route would you choose if you got the branch backwards?	Key: Return to condition-result-first tracing; write True/False before naming the branch.

Support route, not scored

Condition result first. Then branch. Then action. Do not choose the action from the scenario label alone.